Krishnaa **Vadivel** | PhD Research Engineer | Scientific Computing | ML | EE / Embedded Systems

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Summary

Electrical engineering PhD researcher and engineer with experience in first-principles materials modeling, semiconductor-device simulation, scientific ML, embedded systems, and performance-oriented software. Combines theory and implementation across Python/C++, CUDA/ROCm, FPGA/MCU platforms, and leadership-class HPC environments.

Experience

Yale University, Electrical Engineering
PhD Research Assistant — *2021–present*

- Develop first-principles and many-body workflows for self-trapped excitons, exciton-phonon coupling, semiconductor-relevant modeling, and reduced effective theories for materials research.
- Build distributed scientific-computing infrastructure in Python and C++ using MPI, OpenMP, PETSc, SLEPc, CUDA, and ROCm for Frontier, Perlmutter, and related HPC systems.
- Investigate ML-assisted acceleration of scientific workflows, including equivariant graph neural networks, molecular-property prediction, and agentic tooling for documentation- and paper-driven input generation.

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer — *2019–2021*

- Developed CUDA and OpenGL tools for waveform processing, visualization, and engineering diagnostics in deployed geological well-logging systems.
- Implemented embedded and digital functionality across Lattice FPGA, PIC32, dsPIC, STM32, and C8051 platforms; supported circuit bring-up, oscilloscope validation, and targeted Altium PCB updates.
- Contributed host-side tooling with C#, ASP-style services, and Microsoft SQL-backed workflows for engineering and field-data operations.

FindLight
Photonics Technical Writer — *2018–2019*

- Wrote technical content on lasers, optics, and photonics devices for engineering-facing audiences.

Selected Projects

Scientific HPC Research: Distributed first-principles and many-body workflows for exciton-phonon problems, materials simulation, and reusable accelerator-enabled compute pipelines.

Semiconductor Device Modeling: Numerical treatments of Poisson, drift-diffusion, and pn-junction-style equations bridging physics, EE, and computational modeling.

Scientific ML and Automation: Equivariant molecular ML, agentic RAG workflows, and structured tooling for documentation-driven scientific input generation.

Embedded and Digital Systems: FPGA/MCU-based engineering work spanning sensor logging, robotics, digital design, and hardware-software integration.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, PowerShell
Scientific / ML: PyTorch, PyTorch Geometric, scikit-learn, NumPy, pandas, SciPy, SymPy, matplotlib
Distributed Compute: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, ROCm
Hardware / Systems: Verilog, FPGA, PIC32, dsPIC, STM32, C8051, Linux, Docker, Kubernetes, gcloud
Domains: Computational physics, semiconductor devices, photonics, embedded systems, scientific software